CPSC 8420: Advanced Machine Learning Fall 2026

Topic 4: Support Vector Machines

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This lecture builds the support vector machine (SVM) from the ground up. We first show that any pair of parallel hyperplanes can be written as $\langle w, x\rangle + b = \pm 1$ and that their distance is $2/\left\|w\right\|_2$. We then develop Lagrangian duality, Slater's condition, and the KKT conditions, and we verify all of them by hand on two one-dimensional problems. With these tools we derive the SVM dual, explain kernels through Mercer's theorem (including the cosine kernel), work through Sequential Minimal Optimization (SMO) with concrete numbers, and finally solve the primal and the dual with MATLAB's `quadprog`. These notes accompany the course slides (`SVM.pdf`) and the CLAIR blog post on SVM.

Notation. Training data are (x_i, y_i), $i = 1, \dots, n$, with $x_i \in \mathbb{R}^d$ and $y_i \in \{+1, -1\}$. (The slides write m for the number of training points.) The classifier is $f(x) = \langle w, x\rangle + b$ and the predicted label is $\operatorname{sign}(f(x))$. All norms are Euclidean unless stated otherwise, and $\mathbf{1}$ is the all-ones vector.

4.1 Hyperplanes, parallel hyperplanes, and the margin

4.1.1 Hyperplanes and their scaling freedom

A hyperplane in $\mathbb{R}^d$ is a set $H = \{x : \langle a, x\rangle = c\}$ with a nonzero normal vector $a \in \mathbb{R}^d$ and an offset $c \in \mathbb{R}$. In $\mathbb{R}^2$ a hyperplane is a line and in $\mathbb{R}^3$ it is a plane. The description is not unique: for any $t \neq 0$,

$$\{x : \langle a, x\rangle = c\} = \{x : \langle ta, x\rangle = tc\}.$$

This scaling freedom is exactly what lets us choose a convenient normalization.

Two hyperplanes are *parallel* if they have parallel normal vectors. After rescaling one of them we may assume they share the same normal a, so two distinct parallel hyperplanes can always be written as

$$H_1 = \{x : \langle a, x\rangle = c_1\}, \qquad H_2 = \{x : \langle a, x\rangle = c_2\}, \qquad a \neq 0,\ c_1 \neq c_2. \tag{4.1}$$

(If $c_1 = c_2$ they would be the same hyperplane.)

4.1.2 The canonical form $\langle w, x\rangle + b = \pm 1$

Theorem 4.1 (Canonical form of two parallel hyperplanes) *Let H_1, H_2 be as in (4.1). Then there is a unique pair $(w, b) \in \mathbb{R}^d \times \mathbb{R}$ such that*

$$H_1 = \{x : \langle w, x\rangle + b = -1\}, \qquad H_2 = \{x : \langle w, x\rangle + b = +1\}.$$

It is given by

$$\boxed{w = \frac{2}{c_2 - c_1}\, a, \qquad b = -\frac{c_1 + c_2}{c_2 - c_1}}. \tag{4.2}$$

Proof. *Existence.* With w, b from (4.2),

$$\langle w, x\rangle + b = \frac{2\,\langle a, x\rangle - (c_1 + c_2)}{c_2 - c_1}.$$

If $x \in H_1$, then $\langle a, x\rangle = c_1$ and the right side equals $(2c_1 - c_1 - c_2)/(c_2 - c_1) = -1$. If $x \in H_2$, it equals $(2c_2 - c_1 - c_2)/(c_2 - c_1) = +1$. Conversely, $\langle w, x\rangle + b = -1$ forces $2\langle a, x\rangle = c_1 + c_2 - (c_2 - c_1) = 2c_1$, so $x \in H_1$; the same argument applies to $+1$ and H_2. Hence the two descriptions define the same sets.

Uniqueness. Suppose (w', b') also works. Since $\{x : \langle w', x\rangle = -1 - b'\} = \{x : \langle a, x\rangle = c_1\}$, the normal vectors must be parallel: $w' = ta$ for some $t \neq 0$. Evaluating the two equations on points of H_1 and H_2 gives $tc_1 + b' = -1$ and $tc_2 + b' = 1$. Subtracting, $t(c_2 - c_1) = 2$, so $t = 2/(c_2 - c_1)$ and then $b' = -1 - tc_1 = -(c_1 + c_2)/(c_2 - c_1)$. These are exactly (4.2). ∎

Which of the two hyperplanes receives the label $+1$ is a choice; swapping the roles replaces (w, b) by $(-w, -b)$. The hyperplane halfway between them is $\{x : \langle w, x\rangle + b = 0\}$, because the value 0 is the average of -1 and $+1$.

4.1.3 The distance between them is $2/\|w\|_2$

Theorem 4.2 (Width of the slab) *The distance between* $\{x : \langle w, x\rangle + b = -1\}$ *and* $\{x : \langle w, x\rangle + b = 1\}$ *is*

$$\boxed{\text{dist} = \frac{2}{\|w\|_2}}.$$

Proof. Pick any point x_- on the first hyperplane and move along the unit normal $u = w/\|w\|_2$: let $x(t) = x_- + tu$. Then

$$\langle w, x(t)\rangle + b = \underbrace{\langle w, x_-\rangle + b}_{=-1} + t\,\frac{\langle w, w\rangle}{\|w\|_2} = -1 + t\,\|w\|_2 .$$

This equals $+1$ exactly when $t = 2/\|w\|_2$, so the point $x_+ = x_- + \frac{2}{\|w\|_2}u$ lies on the second hyperplane. Moving along the normal is the shortest way between two parallel hyperplanes: for any x'_+ on the second hyperplane, $x'_+ - x_-$ has normal component $\langle u, x'_+ - x_-\rangle = \big(\langle w, x'_+\rangle - \langle w, x_-\rangle\big)/\|w\|_2 = 2/\|w\|_2$, so $\|x'_+ - x_-\|_2 \geq |\langle u, x'_+ - x_-\rangle| = 2/\|w\|_2$ by Cauchy–Schwarz, with equality for $x'_+ = x_+$. ∎

The same argument with the general description (4.1) gives $\text{dist}(H_1, H_2) = |c_2 - c_1|/\|a\|_2$. This agrees with the theorem, since $\|w\|_2 = 2\|a\|_2/|c_2 - c_1|$ by (4.2).

4.1.4 A concrete example with two lines

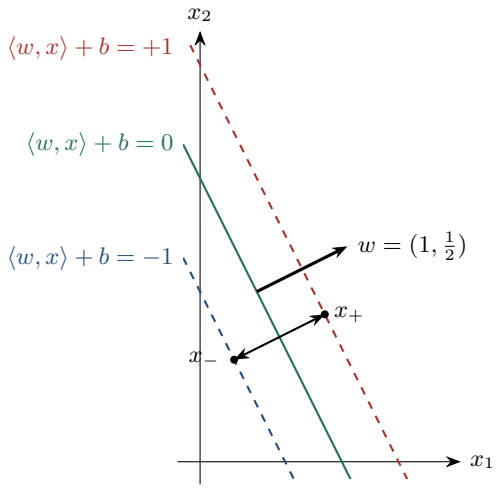
Take the parallel lines $2x_1 + x_2 = 3$ and $2x_1 + x_2 = 7$ in $\mathbb{R}^2$, so $a = (2, 1)^\top$, $c_1 = 3$, $c_2 = 7$. Formula (4.2) gives

$$w = \frac{2}{7 - 3}\begin{pmatrix}2\\1\end{pmatrix} = \begin{pmatrix}1\\1/2\end{pmatrix}, \qquad b = -\frac{3 + 7}{7 - 3} = -\frac{5}{2}.$$

Check: on the first line $\langle w, x\rangle + b = \frac{1}{2}(2x_1 + x_2) - \frac{5}{2} = \frac{3}{2} - \frac{5}{2} = -1$, and on the second line it is $\frac{7}{2} - \frac{5}{2} = 1$. The middle line is $2x_1 + x_2 = 5$. The width is

$$\frac{2}{\|w\|_2} = \frac{2}{\sqrt{1 + 1/4}} = \frac{4}{\sqrt{5}} \approx 1.7889 = \frac{|7 - 3|}{\|(2, 1)\|_2}.$$

For instance, the point $x_- = (0.6, 1.8)$ lies on the first line, and $x_+ = x_- + \frac{4}{\sqrt{5}} \cdot \frac{(2,1)}{\sqrt{5}} = (2.2, 2.6)$ lies on the second line, since $2(2.2) + 2.6 = 7$.



The double arrow from x_- to x_+ has length $2/\|w\|_2 = 4/\sqrt{5}$. The axes use equal scales, so it is drawn perpendicular to the lines.

4.2 The maximum-margin classifier and the soft margin

4.2.1 Hard margin

Suppose the data are linearly separable: some (w, b) has $y_i(\langle w, x_i\rangle + b) > 0$ for all i. The distance from x_i to the decision hyperplane $\{x : \langle w, x\rangle + b = 0\}$ is $|\langle w, x_i\rangle + b|/\|w\|_2$, which for a correctly classified point equals $y_i(\langle w, x_i\rangle + b)/\|w\|_2$. Because (w, b) and (tw, tb) describe the same hyperplane for any $t > 0$, we may fix the scale so that the closest points satisfy $y_i(\langle w, x_i\rangle + b) = 1$. These points lie on the two hyperplanes $\langle w, x\rangle + b = \pm 1$ of Section 4.1, and every point satisfies $y_i(\langle w, x_i\rangle + b) \geq 1$. By Theorem 4.2 the margin width is $2/\|w\|_2$. Maximizing it is equivalent to minimizing $\frac{1}{2}\|w\|_2^2$:

$$\boxed{\min_{w,b} \frac{1}{2}\|w\|_2^2 \quad \text{s.t.} \quad y_i(\langle w, x_i\rangle + b) \geq 1, \quad i = 1, \ldots, n.} \tag{4.3}$$

The objective is a convex quadratic and the constraints are affine in (w, b), so (4.3) is a convex quadratic program (QP).

4.2.2 Soft margin

If the data are not separable, (4.3) has no feasible point. The soft-margin SVM introduces a slack $\xi_i \geq 0$ per point and pays $C\xi_i$ for it, where $C > 0$ trades off margin width against violations:

$$\boxed{\min_{w,b,\xi} \frac{1}{2}\|w\|_2^2 + C\sum_{i=1}^{n} \xi_i \quad \text{s.t.} \quad y_i(\langle w, x_i\rangle + b) \geq 1 - \xi_i, \quad \xi_i \geq 0.} \tag{4.4}$$

The value of ξ_i says where x_i sits: $\xi_i = 0$ means on or outside its margin, $0 < \xi_i < 1$ means inside the margin but correctly classified, $\xi_i = 1$ means on the decision boundary, and $\xi_i > 1$ means misclassified. At the optimum $\xi_i = \max\{0, 1 - y_i(\langle w, x_i\rangle + b)\}$, so (4.4) is the same as minimizing $\frac{1}{2}\|w\|_2^2 + C\sum_i \max\{0, 1 - y_i f(x_i)\}$, the hinge-loss form.

4.3 Lagrangian duality

4.3.1 The primal problem and the Lagrangian

We write constraints in the "≥ 0" form that the SVM uses:

$$p^\star = \min_x \; f(x) \quad \text{s.t.} \quad c_i(x) \geq 0 \; (i = 1, \dots, m), \qquad h_j(x) = 0 \; (j = 1, \dots, p). \tag{4.5}$$

The *Lagrangian* attaches a multiplier to each constraint:

$$L(x, \alpha, \nu) = f(x) - \sum_{i=1}^{m} \alpha_i c_i(x) + \sum_{j=1}^{p} \nu_j h_j(x), \qquad \alpha_i \geq 0, \quad \nu_j \in \mathbb{R}. \tag{4.6}$$

The minus sign in front of α_i matches the "≥ 0" constraints. The slides' generalized Lagrangian uses $g_i(x) \leq 0$ and $+\alpha_i g_i(x)$; the two are identical with $g_i = -c_i$.

The primal as a min–max. For fixed x,

$$\max_{\alpha \geq 0, \nu} L(x, \alpha, \nu) = \begin{cases} f(x), & x \text{ feasible}, \\ +\infty, & \text{otherwise}. \end{cases}$$

Indeed, if some $c_i(x) < 0$ we let $\alpha_i \to \infty$; if some $h_j(x) \neq 0$ we let $\nu_j h_j(x) \to +\infty$; and if x is feasible every penalty term is ≤ 0, so the maximum is reached at $\alpha = 0$. Therefore

$$p^\star = \min_x \max_{\alpha \geq 0, \nu} L(x, \alpha, \nu).$$

The dual problem swaps the order of min and max.

4.3.2 The dual function and the dual problem

Definition 4.3 (Dual function and dual problem) *The dual function is $q(\alpha, \nu) = \inf_x L(x, \alpha, \nu)$ and the dual problem is*

$$d^\star = \max_{\alpha \geq 0, \; \nu} \; q(\alpha, \nu) = \max_{\alpha \geq 0, \nu} \; \min_x \; L(x, \alpha, \nu).$$

For each fixed x, $L(x, \alpha, \nu)$ is affine in (α, ν). A pointwise infimum of affine functions is concave, so q is concave *even when the primal is not convex*. The dual is therefore always a concave maximization.

4.3.3 Weak duality

Theorem 4.4 (Weak duality) *For every primal-feasible x and every $\alpha \geq 0$, ν, $q(\alpha, \nu) \leq f(x)$. Consequently $d^\star \leq p^\star$.*

Proof. If x is feasible and $\alpha \geq 0$, then $\alpha_i c_i(x) \geq 0$ and $h_j(x) = 0$, so $L(x, \alpha, \nu) = f(x) - \sum_i \alpha_i c_i(x) \leq f(x)$. Hence $q(\alpha, \nu) = \inf_z L(z, \alpha, \nu) \leq L(x, \alpha, \nu) \leq f(x)$. Take the supremum over (α, ν) on the left and the infimum over feasible x on the right. ∎

The nonnegative number $p^\star - d^\star$ is the *duality gap*. When it is zero we say that *strong duality* holds.

4.4 Slater's condition and strong duality

Strong duality does not hold in general. For convex problems, a mild regularity condition guarantees it.

Theorem 4.5 (Slater's condition) *Suppose problem* (4.5) *is convex: f is convex, every c_i is concave (so $\{c_i \geq 0\}$ is a convex set), and every h_j is affine. If there is a* strictly feasible *point $\tilde{x}$ in the relative interior of the domain,*

$$c_i(\tilde{x}) > 0 \quad \text{for all } i, \qquad h_j(\tilde{x}) = 0 \quad \text{for all } j,$$

then strong duality holds, $p^\star = d^\star$, and if $p^\star$ is finite the dual optimum is attained by some $(\alpha^\star, \nu^\star)$.

Refined Slater condition. Strict inequality is only needed for the constraints c_i that are not affine. In particular, if every constraint is affine (as in every QP in this lecture), *feasibility alone* implies strong duality whenever the problem is convex.

Why the condition matters. Consider $\min x$ subject to $c(x) = -x^2 \geq 0$. The feasible set is $\{0\}$, so $p^\star = 0$, but no point satisfies $-x^2 > 0$: Slater fails. Here $L(x, \alpha) = x + \alpha x^2$, so $q(\alpha) = -1/(4\alpha)$ for $\alpha > 0$ and $q(0) = -\infty$. Thus $d^\star = \sup_{\alpha \geq 0} q(\alpha) = 0$, but no α attains it. The KKT conditions of Section 4.5 also fail: stationarity $1 + 2\alpha x = 0$ is impossible at $x^\star = 0$. Slater's condition rules out such degenerate cases.

4.5 The Karush–Kuhn–Tucker conditions

Assume f, c_i, h_j are differentiable.

Definition 4.6 (KKT conditions) *A triple $(x^\star, \alpha^\star, \nu^\star)$ satisfies the KKT conditions if*

$$\text{(stationarity)} \qquad \nabla f(x^\star) - \sum_i \alpha_i^\star \nabla c_i(x^\star) + \sum_j \nu_j^\star \nabla h_j(x^\star) = 0, \tag{4.7}$$

$$\text{(primal feasibility)} \qquad c_i(x^\star) \geq 0, \quad h_j(x^\star) = 0, \tag{4.8}$$

$$\text{(dual feasibility)} \qquad \alpha_i^\star \geq 0, \tag{4.9}$$

$$\text{(complementary slackness)} \qquad \alpha_i^\star \, c_i(x^\star) = 0 \quad \text{for every } i. \tag{4.10}$$

Complementary slackness says that for each i, either the multiplier is zero or the constraint is active ($c_i(x^\star) = 0$), or both. An inactive constraint ($c_i(x^\star) > 0$) must have $\alpha_i^\star = 0$.

Theorem 4.7 (KKT and strong duality) *Consider problem* (4.5) *with differentiable f, c_i, h_j.*

1. *(Necessity) If $x^\star$ is primal optimal, $(\alpha^\star, \nu^\star)$ is dual optimal, and $p^\star = d^\star$, then the KKT conditions hold.*
2. *(Sufficiency) If the problem is convex and $(x^\star, \alpha^\star, \nu^\star)$ satisfies the KKT conditions, then $x^\star$ is primal optimal, $(\alpha^\star, \nu^\star)$ is dual optimal, and $p^\star = d^\star$.*

Consequently, for a convex problem satisfying Slater's condition, $x^\star$ is optimal if and only if there exist multipliers satisfying the KKT conditions.

Proof. *Necessity.* Primal and dual feasibility hold by assumption. Using $p^\star = d^\star$,

$$f(x^\star) = q(\alpha^\star, \nu^\star) = \inf_x L(x, \alpha^\star, \nu^\star) \leq L(x^\star, \alpha^\star, \nu^\star) = f(x^\star) - \sum_i \alpha_i^\star c_i(x^\star) \leq f(x^\star).$$

Both inequalities are therefore equalities. The second gives $\sum_i \alpha_i^\star c_i(x^\star) = 0$; since every term is nonnegative, each term is zero, which is (4.10). The first says that $x^\star$ minimizes the differentiable function $L(\cdot, \alpha^\star, \nu^\star)$, so its gradient vanishes there, which is (4.7).

Sufficiency. With $\alpha^\star \geq 0$, convex f, concave c_i, and affine h_j, the function $L(\cdot, \alpha^\star, \nu^\star)$ is convex. Stationarity then makes $x^\star$ a global minimizer of it, so $q(\alpha^\star, \nu^\star) = L(x^\star, \alpha^\star, \nu^\star) = f(x^\star)$ by complementary slackness and $h_j(x^\star) = 0$. Weak duality now forces $f(x^\star) = p^\star = d^\star = q(\alpha^\star, \nu^\star)$. ∎

4.5.1 Recovering the primal solution from the dual

The necessity proof shows more: under strong duality, *every* primal optimum minimizes $L(\cdot, \alpha^\star, \nu^\star)$. Hence

$$\text{if } x \mapsto L(x, \alpha^\star, \nu^\star) \text{ has a unique minimizer } \hat{x}, \text{ then } \hat{x} = x^\star. \tag{4.11}$$

This is how we solve the SVM: we solve the dual for $\alpha^\star$, recover $w^\star$ from stationarity, and recover $b^\star$ from complementary slackness.

4.6 Worked example: $\frac{1}{2}x^2 - x + 1$ with one constraint

Let $f(x) = \frac{1}{2}x^2 - x + 1$. Completing the square,

$$f(x) = \frac{1}{2}(x-1)^2 + \frac{1}{2},$$

so the unconstrained minimizer is $x = 1$ with value $\frac{1}{2}$. We study two constrained problems:

$$\text{(A)} \quad \min_x f(x) \ \text{ s.t. } \ x - 2 \geq 0, \qquad\qquad \text{(B)} \quad \min_x f(x) \ \text{ s.t. } \ x - 1 \geq 0.$$

4.6.1 Problem (A): the constraint $x - 2 \geq 0$

Primal. On $[2, \infty)$, $f'(x) = x - 1 > 0$, so f is increasing and the minimum is at the boundary: $x^\star = 2$ and $p^\star = f(2) = 2 - 2 + 1 = 1$.

Slater. f is convex ($f'' = 1 > 0$) and the constraint is affine. The point $\tilde{x} = 3$ satisfies $3 - 2 = 1 > 0$, so Slater's condition holds and we expect $p^\star = d^\star$ with an attained dual optimum.

Lagrangian and dual function. With $\alpha \geq 0$,

$$L(x, \alpha) = \frac{1}{2}x^2 - x + 1 - \alpha(x - 2).$$

L is a strictly convex quadratic in x; setting $\partial L / \partial x = x - 1 - \alpha = 0$ gives the unique minimizer

$$x(\alpha) = 1 + \alpha. \tag{4.12}$$

Substituting,

$$\begin{aligned} q_A(\alpha) &= \frac{1}{2}(1+\alpha)^2 - (1+\alpha) + 1 - \alpha(1 + \alpha - 2) \\ &= \Big(\frac{1}{2} + \alpha + \frac{1}{2}\alpha^2\Big) - 1 - \alpha + 1 - \alpha^2 + \alpha = -\frac{1}{2}\alpha^2 + \alpha + \frac{1}{2}. \end{aligned}$$

Dual problem. Maximize the concave q_A over $\alpha \geq 0$: $q_A'(\alpha) = 1 - \alpha = 0$ gives $\alpha^\star = 1 \geq 0$, and

$$d^\star = q_A(1) = -\frac{1}{2} + 1 + \frac{1}{2} = 1 = p^\star.$$

Strong duality holds, as Slater predicted.

KKT check at $(x^\star, \alpha^\star) = (2, 1)$:

- stationarity: $x^\star - 1 - \alpha^\star = 2 - 1 - 1 = 0$;
- primal feasibility: $x^\star - 2 = 0 \geq 0$;
- dual feasibility: $\alpha^\star = 1 \geq 0$;
- complementary slackness: $\alpha^\star(x^\star - 2) = 1 \cdot 0 = 0$.

The constraint is active and its multiplier is positive: the constraint is genuinely "pushing" the solution away from the unconstrained minimum $x = 1$.

Recovering the primal from the dual. By (4.11) and the unique minimizer (4.12), $x^\star = x(\alpha^\star) = 1+1 = 2$. We recover the primal solution without ever solving the primal directly.

4.6.2 Problem (B): the constraint $x - 1 \geq 0$

Primal. The unconstrained minimizer $x = 1$ is feasible, so $x^\star = 1$ and $p^\star = f(1) = \frac{1}{2}$.

Slater. $\tilde{x} = 2$ gives $2 - 1 = 1 > 0$, so Slater's condition holds.

Lagrangian and dual function. Now $L(x, \alpha) = \frac{1}{2}x^2 - x + 1 - \alpha(x - 1)$. Again $\partial L/\partial x = x - 1 - \alpha = 0$, so $x(\alpha) = 1 + \alpha$, and

$$q_B(\alpha) = \frac{1}{2}(1+\alpha)^2 - (1+\alpha) + 1 - \alpha \cdot \alpha = \frac{1}{2} + \alpha + \frac{1}{2}\alpha^2 - \alpha - \alpha^2 = \frac{1}{2} - \frac{1}{2}\alpha^2.$$

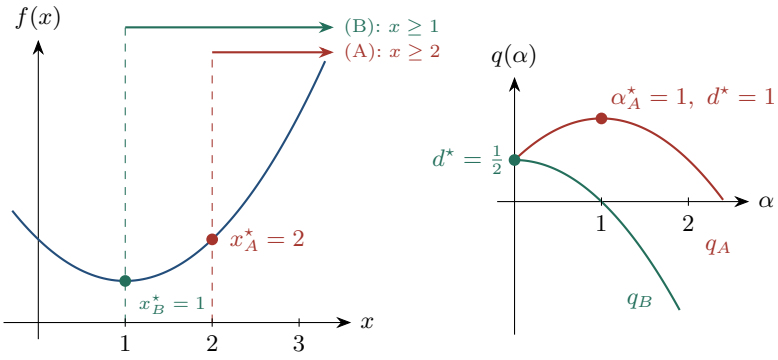
Dual problem. q_B is decreasing on $\alpha \geq 0$, so $\alpha^\star = 0$ and $d^\star = q_B(0) = \frac{1}{2} = p^\star$. Strong duality holds.

KKT check at $(x^\star, \alpha^\star) = (1, 0)$:

- stationarity: $x^\star - 1 - \alpha^\star = 1 - 1 - 0 = 0$;
- primal feasibility: $x^\star - 1 = 0 \geq 0$;
- dual feasibility: $\alpha^\star = 0 \geq 0$;
- complementary slackness: $\alpha^\star(x^\star - 1) = 0 \cdot 0 = 0$.

Here both factors in complementary slackness are zero. The constraint is active (the solution sits on the boundary $x = 1$), but it is not needed: removing it would not change the solution, so its "price" $\alpha^\star$ is zero. This shows that complementary slackness only requires *at least one* factor to vanish. If we had used $x \geq 0$ instead, the constraint would be inactive ($x^\star - 0 = 1 > 0$) and complementary slackness would force $\alpha^\star = 0$.

Recovering the primal. $x^\star = x(\alpha^\star) = 1 + 0 = 1$.



Problem	$x^\star$	$p^\star$	$q(\alpha)$	$\alpha^\star$	$d^\star$	$c(x^\star)$
(A) $x - 2 \geq 0$	2	1	$-\frac{1}{2}\alpha^2 + \alpha + \frac{1}{2}$	1	1	0 (active)
(B) $x - 1 \geq 0$	1	$\frac{1}{2}$	$\frac{1}{2} - \frac{1}{2}\alpha^2$	0	$\frac{1}{2}$	0 (active, unneeded)

4.7 The SVM dual

4.7.1 Hard-margin dual

Apply Section 4.3 to (4.3) with $c_i(w, b) = y_i(\langle w, x_i \rangle + b) - 1 \geq 0$:

$$L(w, b, \alpha) = \frac{1}{2}\|w\|_2^2 - \sum_{i=1}^{n} \alpha_i \big[y_i(\langle w, x_i \rangle + b) - 1\big], \qquad \alpha_i \geq 0. \tag{4.13}$$

For fixed α, minimize over (w, b):

$$\frac{\partial L}{\partial w} = w - \sum_i \alpha_i y_i x_i = 0 \ \Rightarrow\ w = \sum_{i=1}^{n} \alpha_i y_i x_i, \qquad \frac{\partial L}{\partial b} = -\sum_i \alpha_i y_i = 0 \ \Rightarrow\ \sum_{i=1}^{n} \alpha_i y_i = 0.$$

If $\sum_i \alpha_i y_i \neq 0$, then L is unbounded below in b and $q(\alpha) = -\infty$, so the condition $\sum_i \alpha_i y_i = 0$ becomes a constraint of the dual. Substituting $w = \sum_j \alpha_j y_j x_j$:

$$q(\alpha) = \frac{1}{2}\Big\|\sum_i \alpha_i y_i x_i\Big\|_2^2 - \Big\langle \sum_j \alpha_j y_j x_j, \sum_i \alpha_i y_i x_i \Big\rangle - b\underbrace{\sum_i \alpha_i y_i}_{=0} + \sum_i \alpha_i$$

$$= \sum_{i=1}^{n} \alpha_i - \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n} \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle .$$

Collect the labels and inner products into the matrix $Q_{ij} = y_i y_j \langle x_i, x_j \rangle$ (the blog post calls it $\mathcal{K}$). Then the hard-margin dual is

$$\max_{\alpha}\ \mathbf{1}^\top \alpha - \frac{1}{2}\alpha^\top Q \alpha \quad \text{s.t.} \quad y^\top \alpha = 0, \quad \alpha \geq 0. \tag{4.14}$$

4.7.2 Soft-margin dual

For (4.4) we need multipliers $\alpha_i \geq 0$ for the margin constraints and $\mu_i \geq 0$ for $\xi_i \geq 0$:

$$L(w, b, \xi, \alpha, \mu) = \frac{1}{2}\|w\|_2^2 + C\sum_i \xi_i - \sum_i \alpha_i \big[y_i(\langle w, x_i \rangle + b) - 1 + \xi_i\big] - \sum_i \mu_i \xi_i. \tag{4.15}$$

Setting the derivatives to zero gives the same $w = \sum_i \alpha_i y_i x_i$ and $\sum_i \alpha_i y_i = 0$, plus

$$\frac{\partial L}{\partial \xi_i} = C - \alpha_i - \mu_i = 0.$$

The terms in ξ then cancel: $C\xi_i - \alpha_i\xi_i - \mu_i\xi_i = 0$. The dual function is the same expression as before, and the only trace of μ is the constraint $\mu_i = C - \alpha_i \geq 0$, that is, $\alpha_i \leq C$. The soft-margin dual is

$$\boxed{\max_{\alpha} \; \mathbf{1}^\top \alpha - \frac{1}{2}\alpha^\top Q \alpha \quad \text{s.t.} \quad y^\top \alpha = 0, \quad 0 \leq \alpha_i \leq C.} \tag{4.16}$$

Equivalently (the form used by `quadprog` and in the slides), $\min_\alpha \frac{1}{2}\alpha^\top Q\alpha - \mathbf{1}^\top\alpha$ under the same constraints. Only the box $0 \leq \alpha \leq C$ distinguishes the soft dual from the hard dual.

4.7.3 Strong duality for the SVM

The SVM primal is a convex QP with affine constraints, so by the refined Slater condition, feasibility is enough. We can also exhibit a strictly feasible point directly:

- **Soft margin.** Take $w = 0$, $b = 0$, $\xi_i = 2$. Then $y_i(\langle 0, x_i\rangle + 0) = 0 > 1 - 2 = -1$ and $\xi_i = 2 > 0$ for every i. This is strictly feasible for *any* data set.
- **Hard margin.** If (w, b) separates the data with $y_i(\langle w, x_i\rangle + b) \geq 1$, then $(2w, 2b)$ gives $y_i(\langle 2w, x_i\rangle + 2b) \geq 2 > 1$.

Hence $p^\star = d^\star$, a dual optimum $\alpha^\star$ exists, and the KKT conditions characterize the solution. This is why solving the dual solves the SVM.

4.7.4 KKT conditions and support vectors

For the soft-margin problem the KKT conditions read

$$\begin{aligned}
&w = \sum_i \alpha_i y_i x_i, \qquad \sum_i \alpha_i y_i = 0, \qquad C - \alpha_i - \mu_i = 0,\\
&y_i(\langle w, x_i\rangle + b) \geq 1 - \xi_i, \qquad \xi_i \geq 0, \qquad \alpha_i \geq 0, \qquad \mu_i \geq 0,\\
&\alpha_i\big[y_i(\langle w, x_i\rangle + b) - 1 + \xi_i\big] = 0, \qquad \mu_i\xi_i = 0.
\end{aligned}$$

Combining them with $\mu_i = C - \alpha_i$ gives three cases:

Multiplier	Consequence	Geometry
$\alpha_i = 0$	$\mu_i = C > 0 \Rightarrow \xi_i = 0$, so $y_i f(x_i) \geq 1$	on or outside the margin
$0 < \alpha_i < C$	$\mu_i > 0 \Rightarrow \xi_i = 0$ and $y_i f(x_i) = 1$	exactly on the margin (free SV)
$\alpha_i = C$	$\mu_i = 0$, $\xi_i = 1 - y_i f(x_i) \geq 0$	inside the margin or misclassified

Points with $\alpha_i > 0$ are the *support vectors*. Only they appear in $w = \sum_i \alpha_i y_i x_i$; deleting any other point does not change the solution.

4.7.5 Recovering w and b from $\alpha^\star$

- $w^\star = \sum_i \alpha_i^\star y_i x_i$. The Lagrangian is strictly convex in w, so by (4.11) this is the unique optimal w.
- The Lagrangian is linear in b, so b is *not* determined by stationarity. Use complementary slackness instead: for any free support vector ($0 < \alpha_s^\star < C$), $y_s(\langle w^\star, x_s\rangle + b^\star) = 1$, and since $y_s^2 = 1$,

$$b^\star = y_s - \langle w^\star, x_s\rangle = y_s - \sum_j \alpha_j^\star y_j \langle x_j, x_s\rangle . \tag{4.17}$$

  In floating point, average (4.17) over all free support vectors. (If no free support vector exists, $b^\star$ is any value in an interval determined by the other KKT inequalities.)

The resulting classifier is

$$f(x) = \sum_{i=1}^{n} \alpha_i^\star y_i \langle x_i, x\rangle + b^\star, \qquad \hat{y} = \operatorname{sign} f(x).$$

4.8 Kernels and Mercer's theorem

4.8.1 The kernel trick

The dual (4.16) and the classifier use the data only through inner products $\langle x_i, x_j\rangle$. If we first map the data by a feature map $\phi : \mathbb{R}^d \to \mathcal{H}$ (possibly very high- or infinite-dimensional), every inner product becomes $\langle \phi(x_i), \phi(x_j)\rangle$. A *kernel* computes it directly:

$$K(x, z) = \langle \phi(x), \phi(z)\rangle .$$

The kernel SVM dual and classifier are

$$\max_\alpha \ \mathbf{1}^\top \alpha - \frac{1}{2}\alpha^\top Q\alpha, \quad Q_{ij} = y_i y_j K(x_i, x_j), \quad y^\top \alpha = 0,\ 0 \le \alpha \le C; \qquad f(x) = \sum_i \alpha_i y_i K(x_i, x) + b. \tag{4.18}$$

For example (see the slides), $\phi(x) = (x_1^2, \sqrt{2}x_1x_2, x_2^2)$ gives $\langle \phi(x), \phi(z)\rangle = (x_1z_1 + x_2z_2)^2 = \langle x, z\rangle^2$, computed in the original 2-D space. Section 4.11 explains in detail how the classifier in (4.18) labels a new test point without ever forming w.

4.8.2 Which functions are valid kernels?

Definition 4.8 (Valid kernel and Gram matrix) *A function $K : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is a* valid *(proper) kernel if there is an inner-product space $\mathcal{H}$ and a map $\phi : \mathcal{X} \to \mathcal{H}$ with $K(x, z) = \langle \phi(x), \phi(z)\rangle$ for all x, z. For points $x_1, \ldots, x_n$, the* Gram (kernel) matrix *is $\mathbf{K} \in \mathbb{R}^{n\times n}$ with $\mathbf{K}_{ij} = K(x_i, x_j)$.*

Theorem 4.9 (Mercer's condition: valid kernel ⇔ SPSD Gram matrices) *A function K is a valid kernel if and only if K is symmetric and, for every n and every choice of points $x_1, \ldots, x_n \in \mathcal{X}$, the Gram matrix $\mathbf{K}$ is symmetric positive semidefinite (SPSD).*

Proof. *($\Rightarrow$) Valid kernel implies SPSD.* Symmetry follows from $\langle \phi(x), \phi(z)\rangle = \langle \phi(z), \phi(x)\rangle$. For any $v \in \mathbb{R}^n$,

$$v^\top \mathbf{K} v = \sum_{i,j} v_i v_j \langle \phi(x_i), \phi(x_j)\rangle = \Big\langle \sum_i v_i \phi(x_i), \sum_j v_j \phi(x_j) \Big\rangle = \Big\| \sum_i v_i \phi(x_i) \Big\|^2 \ge 0.$$

(⇐) SPSD implies valid, for a finite domain. Let $\mathcal{X} = \{x_1, \dots, x_N\}$ and let $\mathbf{K} \succeq 0$ be its Gram matrix. By the spectral theorem, $\mathbf{K} = U\Lambda U^\top$ with U orthogonal and $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_N)$, $\lambda_k \geq 0$. Define

$$\phi(x_i) = \Lambda^{1/2} U^\top e_i \in \mathbb{R}^N,$$

which uses $\sqrt{\lambda_k}$, real because $\lambda_k \geq 0$. Then $\langle \phi(x_i), \phi(x_j) \rangle = e_i^\top U \Lambda^{1/2} \Lambda^{1/2} U^\top e_j = \mathbf{K}_{ij} = K(x_i, x_j)$. ∎

Remarks. (i) For an infinite domain the (⇐) direction still holds: the feature space is the reproducing kernel Hilbert space built from the functions $K(\cdot, x)$ (the Moore–Aronszajn construction), with $\phi(x) = K(\cdot, x)$. (ii) Mercer's original 1909 theorem is the continuous version: if K is continuous and symmetric on a compact set and $\iint K(x,z) g(x) g(z)\, dx\, dz \geq 0$ for all square-integrable g, then $K(x,z) = \sum_k \lambda_k e_k(x) e_k(z)$ with $\lambda_k \geq 0$, so $\phi(x) = (\sqrt{\lambda_k}\, e_k(x))_k$. In machine learning, "Mercer's condition" usually refers to the finite-sample statement of Theorem 4.9.

Why the SVM needs it. Let $D_y = \operatorname{diag}(y)$. Then $Q = D_y \mathbf{K} D_y$, and for any v, $v^\top Q v = (D_y v)^\top \mathbf{K} (D_y v) \geq 0$. A valid kernel therefore makes $Q \succeq 0$, the dual objective concave, and the dual a convex QP that `quadprog` solves to global optimality. With an invalid "kernel", Q can be indefinite and the QP is no longer convex.

A non-example. The squared distance $K(x,z) = \|x - z\|_2^2$ is *not* a valid kernel. On the two points $x_1 = 0$, $x_2 = 1$ in $\mathbb{R}$ its Gram matrix is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, with eigenvalues 1 and -1; one negative eigenvalue is enough to violate Theorem 4.9.

Closure rules (stated without proof). Positive combinations and products of valid kernels are valid (the product rule is the Schur product theorem). Hence the polynomial kernel $(\langle x, z \rangle + c)^k$ with $c \geq 0$ and the Gaussian kernel $\exp(-\|x - z\|_2^2 / (2\sigma^2))$ of the slides are valid.

4.8.3 The cosine kernel is valid

For nonzero $x, z \in \mathbb{R}^d$, the cosine (similarity) kernel is

$$K_{\cos}(x, z) = \frac{\langle x, z \rangle}{\|x\|_2 \, \|z\|_2} = \cos \theta_{xz},$$

where θ_{xz} is the angle between x and z.

Proposition 4.10 *$K_{\cos}$ is a valid kernel.*

First proof (feature map). Let $\phi(x) = x / \|x\|_2$, the normalized vector. Then $\langle \phi(x), \phi(z) \rangle = \langle x, z \rangle / (\|x\|_2 \, \|z\|_2) = K_{\cos}(x, z)$, so $K_{\cos}$ is an inner product of features by definition. ∎

Second proof (Mercer's condition). Fix nonzero $x_1, \dots, x_n$, let $G_{ij} = \langle x_i, x_j \rangle$ be the linear Gram matrix ($G = XX^\top \succeq 0$ when the rows of X are the x_i), and let $D = \operatorname{diag}(\|x_1\|_2, \dots, \|x_n\|_2)$, which is invertible. Then $\mathbf{K}_{\cos} = D^{-1} G D^{-1}$ is symmetric, and for every $v \in \mathbb{R}^n$, with $u = D^{-1} v$,

$$v^\top \mathbf{K}_{\cos} v = u^\top G u = \Big\| \sum_i u_i x_i \Big\|_2^2 \geq 0.$$

So every Gram matrix is SPSD, and Theorem 4.9 applies. ∎

A numerical check. Take $x_1 = (1, 0)$, $x_2 = (1, 1)$, $x_3 = (0, 1)$, $x_4 = (-1, 1)$, at angles $0°, 45°, 90°, 135°$. With $s = 1/\sqrt{2} \approx 0.7071$,

$$\mathbf{K}_{\cos} = \begin{pmatrix} 1 & s & 0 & -s \\ s & 1 & s & 0 \\ 0 & s & 1 & s \\ -s & 0 & s & 1 \end{pmatrix}.$$

Its eigenvalues are $2, 2, 0, 0$, all nonnegative. To see this exactly, write $\mathbf{K}_{\cos} = \Phi\Phi^\top$ where the rows of $\Phi \in \mathbb{R}^{4\times 2}$ are the unit vectors $\phi(x_i) = (\cos\theta_i, \sin\theta_i)$. The nonzero eigenvalues of $\Phi\Phi^\top$ equal those of

$$\Phi^\top\Phi = \begin{pmatrix} \sum \cos^2\theta_i & \sum \cos\theta_i \sin\theta_i \\ \sum \cos\theta_i \sin\theta_i & \sum \sin^2\theta_i \end{pmatrix} = \begin{pmatrix} 1+\frac{1}{2}+0+\frac{1}{2} & 0+\frac{1}{2}+0-\frac{1}{2} \\ 0+\frac{1}{2}+0-\frac{1}{2} & 0+\frac{1}{2}+1+\frac{1}{2} \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix},$$

and the remaining two eigenvalues are 0 because $\operatorname{rank}\Phi \le 2$.

The shift form $\cos(s-t)$. On scalars $s, t \in \mathbb{R}$, the kernel $K(s,t) = \cos(s-t)$ is also valid, because

$$\cos(s-t) = \cos s\cos t + \sin s \sin t = \langle \phi(s), \phi(t)\rangle, \qquad \phi(s) = (\cos s, \sin s).$$

It is exactly the cosine kernel applied to the points $\phi(s)$ on the unit circle.

4.9 Sequential Minimal Optimization (SMO)

4.9.1 Why two variables at a time

We maximize the dual objective

$$W(\alpha) = \sum_{i=1}^{n} \alpha_i - \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n} \alpha_i\alpha_j y_i y_j K_{ij}, \qquad K_{ij} = K(x_i, x_j),$$

subject to $0 \le \alpha_i \le C$ and $\sum_i \alpha_i y_i = 0$. Coordinate ascent on a single α_i is impossible: with all others fixed, the equality constraint determines α_i completely. SMO therefore updates a *pair* (α_1, α_2) (any two indices; we relabel them $1, 2$) while keeping $\alpha_3, \ldots, \alpha_n$ fixed. The equality constraint becomes

$$\alpha_1 y_1 + \alpha_2 y_2 = \zeta, \qquad \zeta := -\sum_{i=3}^{n} \alpha_i y_i, \tag{4.19}$$

a line in the (α_1, α_2) plane, intersected with the box $[0, C]^2$.

4.9.2 The feasible segment: bounds $L \le \alpha_2 \le H$

Multiply (4.19) by y_1 and use $y_1^2 = 1$: $\alpha_1 = y_1\zeta - y_1 y_2 \alpha_2$.

- If $y_1 \ne y_2$: $\alpha_1 - \alpha_2$ is fixed, the segment has slope $+1$, and

$$L = \max(0,\ \alpha_2^{\text{old}} - \alpha_1^{\text{old}}), \qquad H = \min(C,\ C + \alpha_2^{\text{old}} - \alpha_1^{\text{old}}).$$

- If $y_1 = y_2$: $\alpha_1 + \alpha_2$ is fixed, the segment has slope -1, and

$$L = \max(0,\ \alpha_1^{\text{old}} + \alpha_2^{\text{old}} - C), \qquad H = \min(C,\ \alpha_1^{\text{old}} + \alpha_2^{\text{old}}).$$

The pair can move only if $L \le H$. When SMO starts from a feasible α (for example $\alpha = 0$) and every step keeps (4.19), this always holds, because the old point itself lies on the segment.

4.9.3 The one-dimensional problem and the update

Let $f(x) = \sum_j \alpha_j y_j K(x_j, x) + b$ be the current classifier and $E_i = f(x_i) - y_i$ the current error on x_i. Define

$$\eta = K_{11} + K_{22} - 2K_{12} = \|\phi(x_1) - \phi(x_2)\|^2 \geq 0.$$

Substituting $\alpha_1 = y_1\zeta - y_1 y_2 \alpha_2$ into W leaves a quadratic in α_2 alone, with second derivative $-\eta$. Write $v_i = \sum_{j\geq 3} \alpha_j y_j K_{ij}$ $(i = 1, 2)$ for the contribution of the fixed variables and $s = y_1 y_2$. Differentiating (using $s^2 = 1$ and $y_1 s = y_2$),

$$\frac{dW}{d\alpha_2} = (1 - s) + y_2\zeta(K_{11} - K_{12}) + y_2(v_1 - v_2) - \eta\,\alpha_2.$$

Expanding $E_1 - E_2$ in terms of v_1, v_2 and the old values shows $(1 - s) + y_2\zeta(K_{11} - K_{12}) + y_2(v_1 - v_2) = \eta\,\alpha_2^{\text{old}} + y_2(E_1 - E_2)$. Therefore, when $\eta > 0$, the unconstrained maximizer along the line is

$$\boxed{\alpha_2^{\text{new,unc}} = \alpha_2^{\text{old}} + \frac{y_2(E_1 - E_2)}{\eta}}. \tag{4.20}$$

Then clip it to the segment and move α_1 so that (4.19) still holds:

$$\alpha_2^{\text{new}} = \min\big(H, \max(L, \alpha_2^{\text{new,unc}})\big), \qquad \alpha_1^{\text{new}} = \alpha_1^{\text{old}} + y_1 y_2\big(\alpha_2^{\text{old}} - \alpha_2^{\text{new}}\big). \tag{4.21}$$

Because W is a concave quadratic along the segment, clipping gives the exact maximizer on the segment. (If $\eta = 0$, W is linear along the line and the maximum is at L or H.)

Updating b. With $\Delta_1 = \alpha_1^{\text{new}} - \alpha_1^{\text{old}}$ and $\Delta_2 = \alpha_2^{\text{new}} - \alpha_2^{\text{old}}$, let

$$b_1 = b - E_1 - y_1\Delta_1 K_{11} - y_2\Delta_2 K_{12},$$
$$b_2 = b - E_2 - y_1\Delta_1 K_{12} - y_2\Delta_2 K_{22}.$$

b_1 is the bias that makes $f^{\text{new}}(x_1) = y_1$, and b_2 makes $f^{\text{new}}(x_2) = y_2$. Use b_1 if $0 < \alpha_1^{\text{new}} < C$, else b_2 if $0 < \alpha_2^{\text{new}} < C$, else $(b_1 + b_2)/2$. This is complementary slackness for a free support vector, as in (4.17).

4.9.4 A worked example

Data. Use the kernel $K(x_i, x_j) = \langle x_j, x_i\rangle = i \cdot j$. This is the linear kernel on the one-dimensional points $x_i = i$, so

$$K_{11} = 1, \quad K_{22} = 4, \quad K_{12} = 2, \quad \eta = 1 + 4 - 2 \cdot 2 = 1.$$

Take $n = 6$ points ($x_i = i$, $i = 1, \ldots, 6$), $C = 1$, and the fixed multipliers

$$(\alpha_3, \alpha_4, \alpha_5, \alpha_6) = (1,\ 0.3,\ 0.2,\ 1).$$

Assume the current bias is $b = 0$. Because K has rank one, $\sum_{i,j} \alpha_i\alpha_j y_i y_j K_{ij} = \big(\sum_i \alpha_i y_i x_i\big)^2$, so

$$W(\alpha) = \sum_i \alpha_i - \frac{1}{2}S^2, \qquad S := \sum_{i=1}^{6} \alpha_i y_i\, i, \qquad f(x) = S\,x + b. \tag{4.22}$$

Case 1: $y_1 = 1$, $y_2 = -1$, with $\sum_{i\geq 3} \alpha_i y_i = 0.5$

Let $(y_3, y_4, y_5, y_6) = (+1, +1, +1, -1)$. Then $\sum_{i\geq 3} \alpha_i y_i = 1 + 0.3 + 0.2 - 1 = 0.5$, so by (4.19)

$$\alpha_1 - \alpha_2 = \zeta = -0.5, \qquad \text{i.e.} \qquad \alpha_1 = \alpha_2 - 0.5.$$

Take the current (feasible) pair $\alpha_1^{\text{old}} = 0.1$, $\alpha_2^{\text{old}} = 0.6$.

Bounds. $y_1 \neq y_2$, so $L = \max(0, 0.6 - 0.1) = 0.5$ and $H = \min(1, 1 + 0.5) = 1$.

Errors. The fixed points contribute $\sum_{j\geq 3} \alpha_j y_j\, j = 3+1.2+1.0-6 = -0.8$, so $S^{\text{old}} = 0.1(1)(1)+0.6(-1)(2)-0.8 = -1.9$. Hence

$$f(x_1) = -1.9, \quad E_1 = -1.9 - 1 = -2.9, \qquad f(x_2) = -3.8, \quad E_2 = -3.8 - (-1) = -2.8.$$

Update. By (4.20),

$$\alpha_2^{\text{new,unc}} = 0.6 + \frac{(-1)(-2.9 - (-2.8))}{1} = 0.6 + 0.1 = 0.7 \in [0.5, 1],$$

so no clipping is needed: $\alpha_2^{\text{new}} = 0.7$ and $\alpha_1^{\text{new}} = 0.1 + (1)(-1)(0.6 - 0.7) = 0.2$. Check: $0.2 - 0.7 = -0.5$.

Direct check by substitution. On the line, $\alpha_1 = \alpha_2 - 0.5$, so $\sum_i \alpha_i = (\alpha_2 - 0.5) + \alpha_2 + 2.5 = 2\alpha_2 + 2$ and $S = (\alpha_2 - 0.5) - 2\alpha_2 - 0.8 = -(\alpha_2 + 1.3)$. By (4.22),

$$W(\alpha_2) = 2\alpha_2 + 2 - \frac{1}{2}(\alpha_2 + 1.3)^2, \qquad W'(\alpha_2) = 2 - (\alpha_2 + 1.3) = 0 \iff \alpha_2 = 0.7.$$

The objective increases from $W(0.6) = 3.2 - \frac{1}{2}(1.9)^2 = 1.395$ to $W(0.7) = 3.4 - \frac{1}{2}(2.0)^2 = 1.4$.

Bias. $\Delta_1 = \Delta_2 = 0.1$ and

$$b_1 = 0 - (-2.9) - (1)(0.1)(1) - (-1)(0.1)(2) = 3, \qquad b_2 = 0 - (-2.8) - (1)(0.1)(2) - (-1)(0.1)(4) = 3.$$

Both new multipliers are free, and $b_1 = b_2 = 3$ as they should be. With $S^{\text{new}} = 0.2 - 1.4 - 0.8 = -2$, the new classifier $f(x) = -2x + 3$ gives $f(x_1) = 1 = y_1$ and $f(x_2) = -1 = y_2$: both points now lie exactly on the margin.

Case 2: $y_1 = y_2 = 1$

With $\sum_{i\geq 3} \alpha_i y_i = 0.5$ the pair has no feasible value. Then (4.19) reads $\alpha_1 + \alpha_2 = -0.5$, which no $\alpha_1, \alpha_2 \geq 0$ can satisfy; in the bound formulas, $H = \min(C, \alpha_1 + \alpha_2) = -0.5 < 0 \leq L$. A feasible α with $y_1 = y_2 = 1$ always has $\alpha_1 + \alpha_2 \geq 0$, that is, $\sum_{i\geq 3} \alpha_i y_i \leq 0$.

A feasible version. Flip the labels of points 3–6: $(y_3, y_4, y_5, y_6) = (-1, -1, -1, +1)$, keeping the same multipliers. Now $\sum_{i\geq 3} \alpha_i y_i = -0.5$ and

$$\alpha_1 + \alpha_2 = \zeta = 0.5.$$

Take $\alpha_1^{\text{old}} = 0.3$, $\alpha_2^{\text{old}} = 0.2$ (still $b = 0$).

Bounds. $y_1 = y_2$, so $L = \max(0, 0.5 - 1) = 0$ and $H = \min(1, 0.5) = 0.5$.

Errors. Now $\sum_{j\geq 3} \alpha_j y_j\, j = -3 - 1.2 - 1.0 + 6 = 0.8$ and $S^{\text{old}} = 0.3(1)(1) + 0.2(1)(2) + 0.8 = 1.5$, so

$$f(x_1) = 1.5, \quad E_1 = 0.5, \qquad f(x_2) = 3, \quad E_2 = 2.$$

Update.

$$\alpha_2^{\text{new,unc}} = 0.2 + \frac{(1)(0.5 - 2)}{1} = -1.3 < L = 0 \quad \Longrightarrow \quad \alpha_2^{\text{new}} = 0, \qquad \alpha_1^{\text{new}} = 0.3 + (1)(1)(0.2 - 0) = 0.5.$$

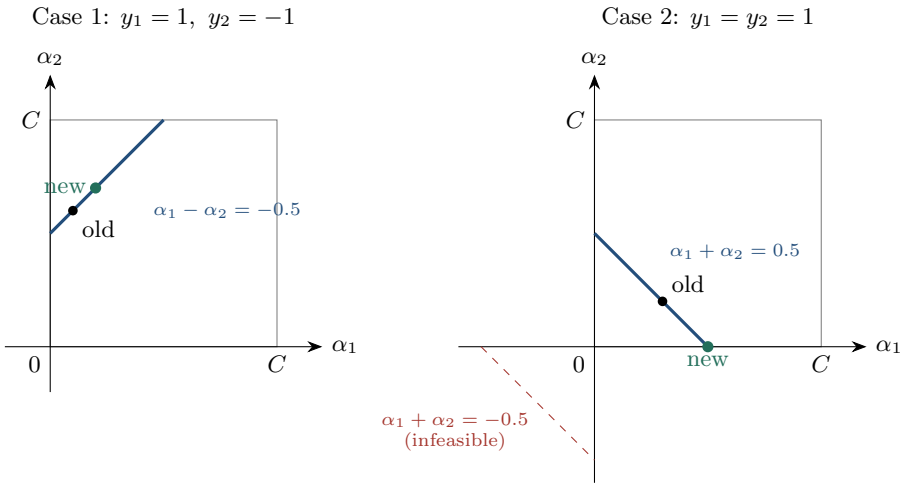
Direct check. On the line, $\alpha_1 = 0.5 - \alpha_2$, so $\sum_i \alpha_i = 0.5 + 2.5 = 3$ is constant and $S = (0.5 - \alpha_2) + 2\alpha_2 + 0.8 = \alpha_2 + 1.3$. Thus $W(\alpha_2) = 3 - \frac{1}{2}(\alpha_2 + 1.3)^2$, whose unconstrained maximizer is $\alpha_2 = -1.3$. On $[0, 0.5]$, W is decreasing, so the maximum is at $\alpha_2 = 0$, confirming the clipped update. The objective increases from

$W(0.2) = 3 - \frac{1}{2}(1.5)^2 = 1.875$ to $W(0) = 3 - \frac{1}{2}(1.3)^2 = 2.155$. Because $y_1 = y_2$, the linear part $\sum_i \alpha_i$ is constant along the line, so this step simply minimizes $\frac{1}{2}S^2 = \frac{1}{2}\|w\|^2$.

Bias. $\Delta_1 = 0.2$, $\Delta_2 = -0.2$:

$$b_1 = 0 - 0.5 - (1)(0.2)(1) - (1)(-0.2)(2) = -0.3, \qquad b_2 = 0 - 2 - (1)(0.2)(2) - (1)(-0.2)(4) = -1.6.$$

Since $0 < \alpha_1^{\text{new}} = 0.5 < C$, we take $b = b_1 = -0.3$. With $S^{\text{new}} = 0.5 + 0 + 0.8 = 1.3$, the classifier $f(x) = 1.3x - 0.3$ gives $f(x_1) = 1 = y_1$ (on the margin, as required for a free multiplier) and $y_2 f(x_2) = 2.3 \geq 1$, consistent with $\alpha_2^{\text{new}} = 0$.



Case	ζ	$[L, H]$	E_1	E_2	$\alpha_2^{\text{new,unc}}$	$(\alpha_1, \alpha_2)^{\text{new}}$	b^{new}	W: old $\to$ new
1: $y_1 \neq y_2$	-0.5	$[0.5, 1]$	-2.9	-2.8	0.7	$(0.2, 0.7)$	3	$1.395 \to 1.4$
2: $y_1 = y_2$	0.5	$[0, 0.5]$	0.5	2	-1.3	$(0.5, 0)$	-0.3	$1.875 \to 2.155$

4.10 Solving the SVM with `quadprog`

4.10.1 The standard form

MATLAB's `quadprog` solves

$$\min_z \ \frac{1}{2} z^\top H z + f^\top z \quad \text{s.t.} \quad Az \leq b, \quad A_{\text{eq}} z = b_{\text{eq}}, \quad lb \leq z \leq ub,$$

called as `z = quadprog(H,f,A,b,Aeq,beq,lb,ub,x0,options)`. Empty brackets `[]` skip a constraint type. Its default algorithm requires a convex problem, that is, $H \succeq 0$; this holds for both SVM formulations.

4.10.2 The data

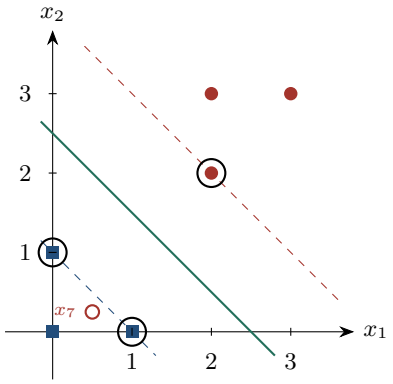
We use six points in $\mathbb{R}^2$ (data set D1):

i	1	2	3	4	5	6
x_i	$(2, 2)$	$(2, 3)$	$(3, 3)$	$(0, 0)$	$(1, 0)$	$(0, 1)$
y_i	$+1$	$+1$	$+1$	-1	-1	-1

The data are separable. The closest points of the two classes are $(2,2)$ and the segment from $(1,0)$ to $(0,1)$, which lie on the lines $x_1+x_2=4$ and $x_1+x_2=1$. By Theorem 4.1 with $a=(1,1)$, $c_1=1$, $c_2=4$, the maximum-margin solution should be

$$w^\star = \frac{2}{3}\begin{pmatrix}1\\1\end{pmatrix}, \qquad b^\star = -\frac{5}{3}, \qquad \text{margin} = \frac{2}{\|w^\star\|_2} = \frac{3}{\sqrt{2}} \approx 2.1213.$$

We now confirm this with `quadprog`, once from the primal and once from the dual, using $C=1$.



Red disks are $y=+1$, blue squares are $y=-1$, and circled points are the support vectors. The solid green line is the decision boundary $x_1+x_2=2.5$, and the dashed lines are the margins $x_1+x_2=1$ (blue) and $x_1+x_2=4$ (red). The hollow point x_7 is used only in Section 4.10.5.

4.10.3 Primal QP

Stack all primal variables into one vector $z=(w;b;\xi)\in\mathbb{R}^{d+1+n}$ (here $d=2$, $n=6$, so $z\in\mathbb{R}^9$). Then

$$\frac{1}{2}\|w\|_2^2 + C\sum_i \xi_i = \frac{1}{2}z^\top H z + f^\top z, \qquad H=\begin{pmatrix} I_d & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0_{n\times n}\end{pmatrix}, \qquad f=\begin{pmatrix}0_{d+1}\\ C\mathbf{1}_n\end{pmatrix}.$$

Each margin constraint $y_i(\langle w,x_i\rangle+b)\ge 1-\xi_i$ is rewritten in the "$\le$" form as

$$-y_i x_i^\top w - y_i b - \xi_i \le -1 \quad\Longleftrightarrow\quad A = -\begin{pmatrix}\operatorname{diag}(y)X & y & I_n\end{pmatrix}, \qquad b_{\text{vec}} = -\mathbf{1}_n,$$

where the rows of X are the $x_i^\top$. The bounds are $\xi\ge 0$, with w and b free. For D1,

$$A = \begin{pmatrix} -2 & -2 & -1 & -1 & 0 & 0 & 0 & 0 & 0\\ -2 & -3 & -1 & 0 & -1 & 0 & 0 & 0 & 0\\ -3 & -3 & -1 & 0 & 0 & -1 & 0 & 0 & 0\\ 0 & 0 & 1 & 0 & 0 & 0 & -1 & 0 & 0\\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & -1 & 0\\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & -1\end{pmatrix}.$$

For instance, row 5 encodes $-(-1)(w_1\cdot 1+w_2\cdot 0)-(-1)b-\xi_5\le -1$, which is $y_5(\langle w,x_5\rangle+b)\ge 1-\xi_5$ for $x_5=(1,0)$, $y_5=-1$.

```
% ---------- Soft-margin SVM: PRIMAL with quadprog ----------
X = [2 2; 2 3; 3 3; 0 0; 1 0; 0 1];   % rows are the points x_i
```

```
y = [1; 1; 1; -1; -1; -1];            % labels (column vector)
C = 1;
[n, d] = size(X);

% variable z = [w; b; xi]  (length d + 1 + n)
H  = blkdiag(eye(d), 0, zeros(n));    % (1/2)||w||^2
f  = [zeros(d+1, 1); C*ones(n, 1)];   % C * sum(xi)
A  = -[diag(y)*X, y, eye(n)];         % y_i(w'x_i + b) >= 1 - xi_i
bv = -ones(n, 1);
lb = [-inf(d+1, 1); zeros(n, 1)];     % w, b free;  xi >= 0
opts = optimoptions('quadprog', 'Display', 'off');

[z, pval] = quadprog(H, f, A, bv, [], [], lb, [], [], opts);
w  = z(1:d);
b  = z(d+1);
xi = z(d+2:end);
fprintf('w = (%.4f, %.4f), b = %.4f, primal value = %.4f\n', w, b, pval);
```

Expected output (up to solver tolerance, about 10^{-8}): $w = (0.6667, 0.6667)$, $b = -1.6667$, $\xi = 0$, and primal value $\frac{1}{2}\|w\|^2 = \frac{4}{9} \approx 0.4444$. This matches the geometric answer. The margins $y_i(\langle w, x_i\rangle + b)$ are $(1, \frac{5}{3}, \frac{7}{3}, \frac{5}{3}, 1, 1)$: points 1, 5, and 6 lie exactly on the margin.

4.10.4 Dual QP

The dual in minimization form is $\min_\alpha \frac{1}{2}\alpha^\top Q\alpha - \mathbf{1}^\top\alpha$ subject to $y^\top\alpha = 0$ and $0 \le \alpha \le C$, so in `quadprog` terms $H = Q$, $f = -\mathbf{1}$, $A_{\text{eq}} = y^\top$, $b_{\text{eq}} = 0$, $lb = 0$, $ub = C\mathbf{1}$. For D1, $\mathbf{K} = XX^\top$ and $Q = (yy^\top) \circ \mathbf{K}$ ($\circ$ is the entrywise product):

$$\mathbf{K} = \begin{pmatrix} 8 & 10 & 12 & 0 & 2 & 2 \\ 10 & 13 & 15 & 0 & 2 & 3 \\ 12 & 15 & 18 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 0 \\ 2 & 3 & 3 & 0 & 0 & 1 \end{pmatrix}, \qquad Q = \begin{pmatrix} 8 & 10 & 12 & 0 & -2 & -2 \\ 10 & 13 & 15 & 0 & -2 & -3 \\ 12 & 15 & 18 & 0 & -3 & -3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -2 & -2 & -3 & 0 & 1 & 0 \\ -2 & -3 & -3 & 0 & 0 & 1 \end{pmatrix}.$$

```
% ---------- Soft-margin SVM: DUAL with quadprog ----------
K   = X*X';                       % linear-kernel Gram matrix
Q   = (y*y').*K;                  % Q_ij = y_i y_j K_ij
Q   = (Q + Q')/2;                 % enforce exact symmetry
f   = -ones(n, 1);
Aeq = y';   beq = 0;              % sum_i alpha_i y_i = 0
lb  = zeros(n, 1);   ub = C*ones(n, 1);

[alpha, dval] = quadprog(Q, f, [], [], Aeq, beq, lb, ub, [], opts);

w   = X'*(alpha.*y);              % stationarity: w = sum_i alpha_i y_i x_i
tol = 1e-6;
sv  = find(alpha > tol);                    % support vectors
fr  = find(alpha > tol & alpha < C - tol);  % free SVs (on the margin)
b   = mean(y(fr) - K(fr,:)*(alpha.*y));     % complementary slackness
fprintf('alpha = %s\n', mat2str(alpha', 4));
fprintf('w = (%.4f, %.4f), b = %.4f, dual value = %.4f\n', w, b, -dval);
```

Expected output. $\alpha = (0.4444,\ 0,\ 0,\ 0,\ 0.2222,\ 0.2222)$, that is, $\alpha^\star = (\frac{4}{9}, 0, 0, 0, \frac{2}{9}, \frac{2}{9})$. The support vectors are points 1, 5, 6, all free since $\alpha_i < C = 1$. Then

$$w = \tfrac{4}{9}(2,2) - \tfrac{2}{9}(1,0) - \tfrac{2}{9}(0,1) = (\tfrac{2}{3}, \tfrac{2}{3}), \qquad b = y_1 - \langle w, x_1\rangle = 1 - \tfrac{8}{3} = -\tfrac{5}{3},$$

and `-dval` $= \mathbf{1}^\top\alpha - \frac{1}{2}\alpha^\top Q\alpha = \frac{8}{9} - \frac{1}{2}\cdot\frac{8}{9} = \frac{4}{9}$. The primal and dual values agree, as strong duality predicts. One can also check $\sum_i \alpha_i y_i = \frac{4}{9} - \frac{2}{9} - \frac{2}{9} = 0$.

Primal or dual? The primal QP has $d+1+n$ variables and the dual has n. The dual is preferable when d is large, and it is the only option when the features $\phi(x)$ are infinite-dimensional (e.g., the Gaussian kernel). The primal is preferable when $n \gg d$.

4.10.5 Adding an outlier: the soft margin at work

Data set D2 adds the point $x_7 = (0.5, 0.25)$ with label $y_7 = +1$. It sits among the negative points, so the data are no longer separable and the hard-margin problem is infeasible. Only the data lines change:

```
X = [2 2; 2 3; 3 3; 0 0; 1 0; 0 1; 0.5 0.25];
y = [1; 1; 1; -1; -1; -1; 1];
```

Re-running both scripts with $C = 1$ gives (exact values in parentheses):

Quantity	Value
w, b	$(0.6667, 0.6667)$, -1.6667 $\quad((\frac{2}{3}, \frac{2}{3}), -\frac{5}{3})$
ξ (primal)	$(0, 0, 0, 0, 0, 0, 2.1667)$ $\quad(\xi_7 = \frac{13}{6})$
α (dual)	$(0.5278, 0, 0, 0, 0.8889, 0.6389, 1)$ $\quad(\frac{19}{36}, 0, 0, 0, \frac{8}{9}, \frac{23}{36}, 1)$
optimal value	2.6111 $\quad(\frac{4}{9} + \frac{13}{6} = \frac{47}{18}$, primal and dual)

Reading the result through KKT. The outlier has $\alpha_7 = C = 1$ and $y_7 f(x_7) = \frac{2}{3}(0.5 + 0.25) - \frac{5}{3} = -\frac{7}{6}$, so $\xi_7 = 1 - (-\frac{7}{6}) = \frac{13}{6} > 1$: it is misclassified, exactly the third case of the KKT table. Points 1, 5, 6 remain free support vectors with $y_i f(x_i) = 1$. Because α_7 is capped at C, the outlier's pull on $w = \sum_i \alpha_i y_i x_i$ is bounded; the other multipliers rise to balance it, and in this example w and b do not change at all. To verify the multipliers by hand, solve

$$2\alpha_1 - \alpha_5 + 0.5 = \tfrac{2}{3}, \qquad 2\alpha_1 - \alpha_6 + 0.25 = \tfrac{2}{3}, \qquad \alpha_1 - \alpha_5 - \alpha_6 + 1 = 0,$$

the two components of $w = \sum_i \alpha_i y_i x_i$ and the constraint $\sum_i \alpha_i y_i = 0$. This gives $\alpha_1 = \frac{19}{36}$, $\alpha_5 = \frac{8}{9}$, $\alpha_6 = \frac{23}{36}$.

4.10.6 A kernel example: XOR with a polynomial kernel

The XOR data $x = (1,1), (-1,-1)$ with $y = +1$ and $x = (1,-1), (-1,1)$ with $y = -1$ cannot be separated by a line. With the polynomial kernel $K(x,z) = (\langle x, z\rangle + 1)^2$ from the slides, only the Gram matrix changes:

```
X = [1 1; -1 -1; 1 -1; -1 1];   y = [1; 1; -1; -1];   C = 1;
n = size(X, 1);                 % opts as defined in the primal script
K = (X*X' + 1).^2;              % polynomial kernel, degree 2, c = 1
Q = (y*y').*K;   Q = (Q + Q')/2;
alpha = quadprog(Q, -ones(n,1), [], [], y', 0, zeros(n,1), C*ones(n,1), [], opts);
fr = find(alpha > 1e-6 & alpha < C - 1e-6);
b  = mean(y(fr) - K(fr,:)*(alpha.*y));
fx = @(x) ((X*x(:) + 1).^2)'*(alpha.*y) + b;   % decision function
```

Here $\mathbf{K}$ has 9 on the diagonal and 1 elsewhere, and $Q = \begin{pmatrix} 9 & 1 & -1 & -1 \\ 1 & 9 & -1 & -1 \\ -1 & -1 & 9 & 1 \\ -1 & -1 & 1 & 9 \end{pmatrix}$, with eigenvalues $12, 8, 8, 8$, so $Q \succ 0$. By symmetry, $\alpha^\star = (\frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8})$ (each $< C$, so all four points are free support vectors), $b^\star = 0$, and the dual value is $\frac{1}{4}$. The decision function simplifies to

$$f(x) = \frac{1}{8}\Big[(x_1 + x_2 + 1)^2 + (-x_1 - x_2 + 1)^2 - (x_1 - x_2 + 1)^2 - (-x_1 + x_2 + 1)^2\Big] = x_1 x_2,$$

whose sign is exactly the XOR rule.

4.11 Classifying a test point with a kernel SVM

4.11.1 Why we never form w

With a feature map ϕ, the primal weight vector lives in the feature space:

$$w^\star = \sum_{i=1}^{n} \alpha_i^\star y_i \, \phi(x_i) \in \mathcal{H}. \tag{4.23}$$

For the linear kernel $\phi(x) = x$, so $w^\star \in \mathbb{R}^d$ can be computed and stored, as in Section 4.10. For a polynomial kernel $\mathcal{H}$ is finite-dimensional but can be huge, and for the Gaussian kernel it is infinite-dimensional, so $w^\star$ cannot be written down as a vector at all. The way out is that a prediction never needs $w^\star$ itself, only its inner product with $\phi(x)$. By linearity of the inner product,

$$\boxed{f(x) = \langle w^\star, \phi(x) \rangle + b^\star = \Big\langle \sum_{i=1}^{n} \alpha_i^\star y_i \phi(x_i),\, \phi(x) \Big\rangle + b^\star = \sum_{i=1}^{n} \alpha_i^\star y_i \, K(x_i, x) + b^\star}, \qquad \hat{y} = \operatorname{sign} f(x). \tag{4.24}$$

Only support vectors ($\alpha_i^\star > 0$) contribute, so the sum runs over the support vectors only. Equation (4.24) has a natural reading: $f(x)$ is a weighted vote of the support vectors, where support vector x_i pushes toward its own label y_i with weight $\alpha_i^\star K(x_i, x)$. For a similarity kernel such as the Gaussian, support vectors close to x have a large $K(x_i, x)$ and dominate the vote.

The bias also needs only kernel values among *training* points. Rewriting (4.17) with (4.23), for any free support vector s ($0 < \alpha_s^\star < C$),

$$b^\star = y_s - \sum_{j=1}^{n} \alpha_j^\star y_j \, K(x_j, x_s), \tag{4.25}$$

averaged over all free support vectors in floating point.

What a trained kernel SVM stores. The support vectors x_i, the coefficients $\alpha_i^\star y_i$, the bias $b^\star$, and the kernel function. Classifying one test point costs one kernel evaluation per support vector.

4.11.2 The recipe, and its matrix form

1. **Train.** Solve the dual (4.18) with the training Gram matrix $\mathbf{K}_{ij} = K(x_i, x_j)$ to get $\alpha^\star$.
2. **Bias.** Compute $b^\star$ from (4.25).
3. **Similarities.** For a test point $\tilde{x}$, compute $K(x_i, \tilde{x})$ for every support vector x_i.

4. **Score.** $f(\tilde{x}) = \sum_i \alpha_i^\star y_i K(x_i, \tilde{x}) + b^\star$.
5. **Label.** $\hat{y} = +1$ if $f(\tilde{x}) > 0$ and $\hat{y} = -1$ if $f(\tilde{x}) < 0$. A test point with $f(\tilde{x}) = 0$ lies exactly on the decision boundary, and any fixed tie rule may be used.

For m test points $\tilde{x}_1, \dots, \tilde{x}_m$, collect the test-versus-training kernel values in $\mathbf{K}_{\text{te}} \in \mathbb{R}^{m \times n}$ with $(\mathbf{K}_{\text{te}})_{ti} = K(\tilde{x}_t, x_i)$. Rows index test points and columns index training points; this is *not* the training Gram matrix. Then

$$f = \mathbf{K}_{\text{te}} (\alpha^\star \circ y) + b^\star \mathbf{1}_m \in \mathbb{R}^m.$$

```
% ---------- Predicting with a trained kernel SVM ----------
% X: n-by-d training points, y: labels in {+1,-1}, alpha and b from the dual.
sigma = 1;
kern  = @(A, B) exp(-(sum(A.^2, 2) + sum(B.^2, 2)' - 2*A*B')/(2*sigma^2));  % Gaussian
% kern = @(A, B) A*B';              % linear kernel
% kern = @(A, B) (A*B' + 1).^2;     % polynomial kernel, degree 2
Kte  = kern(Xtest, X);              % Kte(t,i) = K(xtest_t, x_i), size m-by-n
f    = Kte*(alpha.*y) + b;          % decision values, one per test point
yhat = sign(f);                     % predicted labels (0 means on the boundary)
```

The Gaussian line expands $\|a - z\|_2^2 = \|a\|_2^2 + \|z\|_2^2 - 2\langle a, z \rangle$ for all pairs at once (implicit expansion, MATLAB R2016b or later). Train with the same `kern`: `K = kern(X, X)`.

4.11.3 Example 1: the linear kernel, where both routes are available

For data set D1 we found $\alpha^\star = (\frac{4}{9}, 0, 0, 0, \frac{2}{9}, \frac{2}{9})$, $b^\star = -\frac{5}{3}$, and $w^\star = (\frac{2}{3}, \frac{2}{3})$. Classify $\tilde{x} = (1, 2)$. Only the support vectors $x_1 = (2, 2)$, $x_5 = (1, 0)$, $x_6 = (0, 1)$ contribute:

$$\begin{aligned} f(\tilde{x}) &= \tfrac{4}{9}(+1)\,\langle (2,2), (1,2) \rangle + \tfrac{2}{9}(-1)\,\langle (1,0), (1,2) \rangle + \tfrac{2}{9}(-1)\,\langle (0,1), (1,2) \rangle - \tfrac{5}{3} \\ &= \tfrac{4}{9} \cdot 6 - \tfrac{2}{9} \cdot 1 - \tfrac{2}{9} \cdot 2 - \tfrac{5}{3} = \tfrac{24}{9} - \tfrac{6}{9} - \tfrac{15}{9} = \tfrac{1}{3} > 0. \end{aligned}$$

The primal route gives the same value, $\langle w^\star, \tilde{x} \rangle + b^\star = \frac{2}{3}(1 + 2) - \frac{5}{3} = \frac{1}{3}$, so $\hat{y} = +1$. With the linear kernel the two routes always agree; the kernel route is the one that still works when $w^\star$ cannot be formed.

4.11.4 Example 2: a polynomial kernel, where w lives in $\mathbb{R}^6$

For the XOR example of Section 4.10 with $K(x, z) = (\langle x, z \rangle + 1)^2$ we found $\alpha_i^\star = \frac{1}{8}$ for all four points and $b^\star = 0$. Expanding the kernel shows the feature map:

$$(x_1 z_1 + x_2 z_2 + 1)^2 = x_1^2 z_1^2 + x_2^2 z_2^2 + 2x_1 x_2 z_1 z_2 + 2x_1 z_1 + 2x_2 z_2 + 1 = \langle \phi(x), \phi(z) \rangle,$$

$$\phi(x) = \big(x_1^2,\ x_2^2,\ \sqrt{2}\,x_1 x_2,\ \sqrt{2}\,x_1,\ \sqrt{2}\,x_2,\ 1\big) \in \mathbb{R}^6.$$

The four training points map to

$$\begin{aligned} \phi(1,1) &= (1, 1, \sqrt{2}, \sqrt{2}, \sqrt{2}, 1), & \phi(-1,-1) &= (1, 1, \sqrt{2}, -\sqrt{2}, -\sqrt{2}, 1), \\ \phi(1,-1) &= (1, 1, -\sqrt{2}, \sqrt{2}, -\sqrt{2}, 1), & \phi(-1,1) &= (1, 1, -\sqrt{2}, -\sqrt{2}, \sqrt{2}, 1). \end{aligned}$$

With labels $(+1, +1, -1, -1)$, (4.23) gives

$$w^\star = \tfrac{1}{8}\big[\phi(1,1) + \phi(-1,-1) - \phi(1,-1) - \phi(-1,1)\big] = \tfrac{1}{8}\big(0, 0, 4\sqrt{2}, 0, 0, 0\big) = \big(0, 0, \tfrac{\sqrt{2}}{2}, 0, 0, 0\big),$$

so $\langle w^\star, \phi(x) \rangle + b^\star = \frac{\sqrt{2}}{2} \cdot \sqrt{2}\, x_1 x_2 = x_1 x_2$, matching the formula found earlier. Now classify $\tilde{x} = (0.5, 2)$ by the kernel route, which never uses ϕ:

Support vector x_i	$(1,1)$	$(-1,-1)$	$(1,-1)$	$(-1,1)$
y_i	$+1$	$+1$	-1	-1
$\langle x_i, \tilde{x}\rangle + 1$	3.5	-1.5	-0.5	2.5
$K(x_i,\tilde{x}) = (\langle x_i,\tilde{x}\rangle+1)^2$	12.25	2.25	0.25	6.25

$$f(\tilde{x}) = \tfrac{1}{8}\big(12.25 + 2.25 - 0.25 - 6.25\big) + 0 = \tfrac{1}{8}\cdot 8 = 1 > 0 \quad\Longrightarrow\quad \hat{y} = +1,$$

in agreement with $x_1 x_2 = 0.5 \cdot 2 = 1$.

4.11.5 Example 3: the Gaussian kernel, where w cannot be written down

Keep the XOR data but use $K(x,z) = \exp\big(-\|x-z\|_2^2/(2\sigma^2)\big)$ with $\sigma = 1$ and $C = 10$. Now $\phi(x)$ is infinite-dimensional, so (4.23) is a perfectly valid formula for $w^\star$, but not one we can evaluate as a list of numbers. Everything goes through K.

Training. Same-class corners are at squared distance 8 and neighboring corners at squared distance 4, so the Gram matrix has 1 on the diagonal, $e^{-4} \approx 0.0183$ between opposite corners, and $e^{-2} \approx 0.1353$ between neighboring corners. By symmetry, try $\alpha_i = a$ for all i and $b = 0$. For $x_1 = (1,1)$,

$$y_1 f(x_1) = a\big[K_{11} + K_{12} - K_{13} - K_{14}\big] = a\big(1 + e^{-4} - 2e^{-2}\big) = a\big(1 - e^{-2}\big)^2,$$

and the same holds for every point. Setting $y_i f(x_i) = 1$ gives

$$a = \frac{1}{(1-e^{-2})^2} \approx 1.3375.$$

All KKT conditions hold: $\sum_i \alpha_i y_i = a(1+1-1-1) = 0$; $0 < a < C$; every point lies on its margin with $\xi_i = 0$, as required for a free support vector. By the sufficiency part of Theorem 4.7, this is the optimal solution. (A `quadprog` run gives the same α and $b = 0$.)

Prediction at $\tilde{x} = (0.5, 2)$.

Support vector x_i	$(1,1)$	$(-1,-1)$	$(1,-1)$	$(-1,1)$
y_i	$+1$	$+1$	-1	-1
$\|x_i - \tilde{x}\|_2^2$	1.25	11.25	9.25	3.25
$K(x_i,\tilde{x}) = e^{-\|x_i-\tilde{x}\|^2/2}$	0.5353	0.0036	0.0098	0.1969

$$f(\tilde{x}) = 1.3375\,(0.5353 + 0.0036 - 0.0098 - 0.1969) + 0 = 1.3375 \times 0.3322 \approx 0.4443 > 0 \quad\Longrightarrow\quad \hat{y} = +1.$$

The nearest support vector, $(1,1)$ with label $+1$, has by far the largest similarity and decides the vote. Note that the kernel value 0.5353 is $e^{-1.25/2} = e^{-0.625}$, not $e^{-1.25}$: the factor $2\sigma^2 = 2$ matters.

Summary of the three examples. In all cases the classifier is $f(x) = \sum_i \alpha_i^\star y_i K(x_i, x) + b^\star$. For the linear kernel we may collapse the sum into $w^\star$ first; for the polynomial kernel we could, at the cost of working in $\mathbb{R}^6$; for the Gaussian kernel we must not, and the kernel sum is the only computable form.

4.12 Multiclass SVM: one-vs-all and one-vs-one

The SVM is a binary classifier. With $M \geq 3$ classes and labels $y_i \in \{1, \ldots, M\}$, the two standard strategies combine several binary SVMs. Each binary SVM below is trained exactly as before (primal or dual, any kernel) and produces a score $f(x) = \sum_i \alpha_i y_i K(x_i, x) + b$ as in (4.24).

4.12.1 One-vs-all (one-vs-rest)

Training. For each class $k = 1, \ldots, M$, relabel the data,

$$y_i^{(k)} = \begin{cases} +1, & y_i = k, \\ -1, & y_i \neq k, \end{cases}$$

and train a binary SVM on *all* n points. This gives M scores

$$f_k(x) = \langle w_k, \phi(x) \rangle + b_k = \sum_{i=1}^{n} \alpha_i^{(k)} y_i^{(k)} K(x_i, x) + b_k, \qquad k = 1, \ldots, M.$$

Prediction.

$$\boxed{\hat{y} = \underset{k \in \{1,\ldots,M\}}{\arg\max} \; f_k(x)} \qquad \Big(\text{linear kernel: } \hat{y} = \underset{k}{\arg\max} \; \langle w_k, x \rangle + b_k\Big). \tag{4.26}$$

Why arg max **and not** arg min**.** Classifier k was trained with class k as the *positive* class. A large positive $f_k(x)$ means x lies deep on the class-k side of its boundary; a negative $f_k(x)$ means classifier k says "not class k." The arg min would choose the class whose own classifier rejects x most strongly, which is the opposite of what we want. The arg max also resolves the three situations the M signs alone cannot:

- exactly one $f_k(x) > 0$: that class wins, and the arg max agrees;
- several $f_k(x) > 0$ (several classifiers claim x): the most confident one wins;
- no $f_k(x) > 0$ (every classifier rejects x): the least negative one wins.

Remark on scales. The M classifiers are trained separately, so their scores are not calibrated against each other. One may compare the signed distances $f_k(x)/\|w_k\|$ instead, with $\|w_k\|^2 = \alpha^{(k)\top} Q^{(k)} \alpha^{(k)}$ computable from the kernel. The standard rule, used below, compares the raw scores $f_k(x)$.

Cost. M binary problems, each with all n training points.

4.12.2 One-vs-one

Training. For each pair of classes $k < l$, keep only the points of classes k and l, label class k as $+1$ and class l as -1, and train a binary SVM with score $f_{kl}(x)$. There are $\binom{M}{2} = M(M-1)/2$ classifiers.

Prediction ("Max Wins" voting). Each pairwise classifier casts one vote:

$$\text{vote for } k \text{ if } f_{kl}(x) > 0, \qquad \text{vote for } l \text{ otherwise,}$$

and

$$\boxed{\hat{y} = \underset{k}{\arg\max} \; \#\{\text{votes received by class } k\}}. \tag{4.27}$$

Each class takes part in $M - 1$ contests, so it can receive at most $M - 1$ votes. If two classes tie in votes, a fixed rule is needed; Hsu and Lin (2002)[1], whose one-vs-one scheme is the one implemented in LIBSVM, simply choose the class with the smaller index. With $M = 3$ a three-way tie with one vote each can occur when the pairwise winners form a cycle (1 beats 2, 2 beats 3, 3 beats 1).

Cost. $M(M-1)/2$ binary problems, but each uses only the points of two classes (about $2n/M$ if classes have equal size). Since the cost of solving the QP grows faster than linearly in the number of points, the many small problems are often cheaper in total than the M large problems of one-vs-all.

[1] C.-W. Hsu and C.-J. Lin, "A comparison of methods for multiclass support vector machines," *IEEE Transactions on Neural Networks*, 13(2), 2002.

4.12.3 A worked three-class example

Take $M = 3$ classes with two points each, the linear kernel, and $C = 10$:

Class	1	2	3
Training points	$(0,2),\ (0,3)$	$(-2,0),\ (-3,0)$	$(2,0),\ (3,0)$

Every binary problem below is separable and its optimal multipliers are all below $C = 10$, so each solution is the maximum-margin separator. Each follows from the two margin lines of the problem via Theorem 4.1, and each was confirmed with `quadprog`.

One-vs-all classifiers.

- Class 1 vs rest: the positives are closest to the negatives along x_2; the margins are $x_2 = 2$ (class 1) and $x_2 = 0$ (the rest), so $f_1(x) = x_2 - 1$.
- Class 2 vs rest: the closest opposing pair is $(-2,0)$ and $(0,2)$, with margins $x_1 + x_2 = -2$ and $x_1 + x_2 = 2$, so $f_2(x) = -\frac{1}{2}(x_1 + x_2)$.
- Class 3 vs rest: by mirror symmetry, $f_3(x) = \frac{1}{2}(x_1 - x_2)$.

For instance, $f_2(-2,0) = 1$, $f_2(0,2) = -1$, $f_2(2,0) = -1$, $f_2(-3,0) = 1.5$, and $f_2(0,3) = f_2(3,0) = -1.5$, so every point satisfies its margin constraint. (In the class-1 problem, four negative points lie on the margin $x_2 = 0$, so its multipliers are not unique; w_1 and b_1 are.)

One-vs-one classifiers (class k is $+1$, class l is -1):

$$f_{12}(x) = \tfrac{1}{2}(x_1 + x_2), \qquad f_{13}(x) = \tfrac{1}{2}(x_2 - x_1), \qquad f_{23}(x) = -\tfrac{1}{2}x_1.$$

For instance, $f_{12}(0,2) = 1$ and $f_{12}(-2,0) = -1$; $f_{23}(-2,0) = 1$ and $f_{23}(2,0) = -1$.

Four test points.

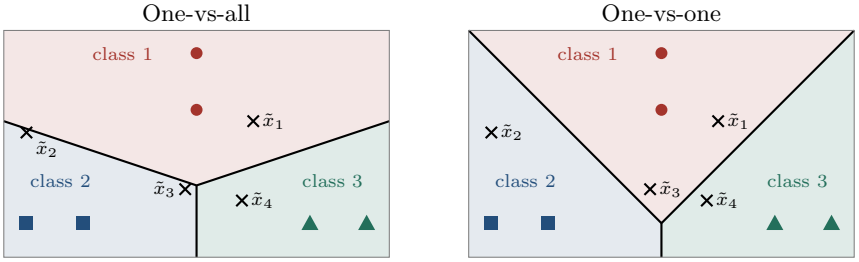
	One-vs-all				Pairwise scores			Votes			One-vs-one
Test point	f_1	f_2	f_3	arg max	f_{12}	f_{13}	f_{23}	1	2	3	winner
$\tilde{x}_1 = (1, 1.8)$	0.8	−1.4	−0.4	1	1.4	0.4	−0.5	2	0	1	1
$\tilde{x}_2 = (-3, 1.6)$	0.6	0.7	−2.3	2	−0.7	2.3	1.5	1	2	0	2
$\tilde{x}_3 = (-0.2, 0.6)$	−0.4	−0.2	−0.4	2	0.2	0.4	0.1	2	1	0	1
$\tilde{x}_4 = (0.8, 0.4)$	−0.6	−0.6	0.2	3	0.6	−0.2	−0.4	1	0	2	3

Reading the table row by row:

- $\tilde{x}_1$: only $f_1 > 0$, so one-vs-all predicts class 1. The pairwise contests give class 1 two wins (over 2 and 3) and class 3 one win (over 2), so one-vs-one also predicts 1. The arg min rule would pick class 2, whose classifier rejects $\tilde{x}_1$ most strongly, even though $\tilde{x}_1$ is far from class 2.
- $\tilde{x}_2$: both f_1 and f_2 are positive; the larger, $f_2 = 0.7$, wins. One-vs-one: class 2 beats class 1 ($f_{12} < 0$) and class 3 ($f_{23} > 0$), so class 2 again.
- $\tilde{x}_3$: no classifier claims the point (all $f_k < 0$); the least negative, $f_2 = -0.2$, wins. One-vs-one gives class 1 two wins, so the two strategies *disagree*. Points like this, near the middle of all three classes, are exactly where the combination rule matters.

- $\tilde{x}_4$: only $f_3 > 0$; one-vs-one gives class 3 two wins. Both predict 3.

Decision regions. Setting the scores equal shows where each rule changes its answer. One-vs-all: $f_2 = f_3$ on $x_1 = 0$, $f_1 = f_2$ on $x_1 + 3x_2 = 2$, and $f_1 = f_3$ on $3x_2 - x_1 = 2$; the three lines meet at $(0, \frac{2}{3})$. One-vs-one: class 1 wins both its contests exactly when $x_2 > |x_1|$, class 2 when $x_1 < 0$ and $x_2 < -x_1$, and class 3 when $x_1 > 0$ and $x_2 < x_1$. The point $\tilde{x}_3 = (-0.2, 0.6)$ satisfies $0.6 > |-0.2|$ but $x_1 + 3x_2 = 1.6 < 2$, which is why the two rules disagree there.



Disks, squares, and triangles are the training points of classes 1, 2, and 3; crosses are the test points. Shading shows the predicted class.

4.12.4 Multiclass SVM with `quadprog`

The binary dual solver from Section 4.10 becomes a reusable function; both strategies then call it repeatedly. In a MATLAB script, local functions must appear at the end of the file.

```
% ---------- Multiclass SVM: one-vs-all and one-vs-one ----------
X    = [0 2; 0 3; -2 0; -3 0; 2 0; 3 0];     % training points (rows)
ylab = [1; 1; 2; 2; 3; 3];                   % class labels
Xte  = [1 1.8; -3 1.6; -0.2 0.6; 0.8 0.4];   % test points (rows)
C    = 10;
kern = @(A, B) A*B';                         % linear kernel; any valid kernel works
classes = unique(ylab);   M = numel(classes);
Ktr = kern(X, X);                            % n-by-n: training vs training
Kte = kern(Xte, X);                          % m-by-n: test (rows) vs training

% One-vs-all: M problems, each on all n points
F = zeros(size(Xte, 1), M);
for k = 1:M
    yk = 2*(ylab == classes(k)) - 1;         % +1 for class k, -1 for the rest
    [a, b] = svm_train_dual(Ktr, yk, C);
    F(:, k) = Kte*(a.*yk) + b;               % scores f_k(x)
end
[~, idx] = max(F, [], 2);   yhat_ova = classes(idx);

% One-vs-one: M(M-1)/2 problems, each on two classes
votes = zeros(size(Xte, 1), M);
for k = 1:M-1
    for l = k+1:M
        S   = (ylab == classes(k)) | (ylab == classes(l));   % classes k and l only
        ykl = 2*(ylab(S) == classes(k)) - 1;                 % +1 for k, -1 for l
        [a, b] = svm_train_dual(Ktr(S, S), ykl, C);
        f = Kte(:, S)*(a.*ykl) + b;                          % scores f_kl(x)
```

```
        votes(:, k) = votes(:, k) + (f > 0);
        votes(:, l) = votes(:, l) + (f <= 0);
    end
end
[~, idx] = max(votes, [], 2);   yhat_ovo = classes(idx);  % ties: smaller index

function [alpha, b] = svm_train_dual(K, y, C)
% Binary soft-margin SVM dual with a precomputed kernel matrix K.
n = numel(y);
Q = (y*y').*K;   Q = (Q + Q')/2;
opts  = optimoptions('quadprog', 'Display', 'off');
alpha = quadprog(Q, -ones(n,1), [], [], y', 0, zeros(n,1), C*ones(n,1), [], opts);
tol = 1e-6;
fr  = find(alpha > tol & alpha < C - tol);   % free support vectors
b   = mean(y(fr) - K(fr, :)*(alpha.*y));     % complementary slackness
end
```

Expected output (up to solver tolerance):

$$F = \begin{pmatrix} 0.8 & -1.4 & -0.4 \\ 0.6 & 0.7 & -2.3 \\ -0.4 & -0.2 & -0.4 \\ -0.6 & -0.6 & 0.2 \end{pmatrix}, \quad \texttt{yhat_ova} = \begin{pmatrix} 1 \\ 2 \\ 2 \\ 3 \end{pmatrix}, \quad \texttt{votes} = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 2 & 0 \\ 2 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix}, \quad \texttt{yhat_ovo} = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 3 \end{pmatrix}.$$

Notes on the code: `Kte(:, S)` keeps the columns of the training points used by the pair (k, l), so the scores of each pairwise classifier use only its own support vectors; MATLAB's `max` returns the first maximum, which implements the smaller-index tie rule; and a test point with $f_{kl} = 0$ is counted for l by the line `(f <= 0)`. To use a kernel, replace `kern` as in Section 4.11; nothing else changes. If a binary problem has no free support vector, `fr` is empty and b must be chosen from the KKT inequalities instead.

4.12.5 Comparing the two strategies

	One-vs-all	One-vs-one
Binary problems	M	$M(M-1)/2$
Points per problem	all n	two classes ($\approx 2n/M$)
Labels in problem k / (k, l)	$+1$ for class k, -1 for all others	$+1$ for k, -1 for l
Prediction	$\arg\max_k f_k(x)$	most votes ("Max Wins")
Ties	equal scores (rare)	equal votes; pick smaller index
Typical issue	raw scores of separately trained classifiers are not calibrated; each problem is imbalanced (one class vs. $M-1$)	many classifiers to store and evaluate at test time

4.13 Summary and review questions

Object	Statement
Canonical hyperplanes	$\langle a, x\rangle = c_{1,2}$ become $\langle w, x\rangle + b = \mp 1$ with $w = \frac{2a}{c_2 - c_1}$, $b = -\frac{c_1 + c_2}{c_2 - c_1}$; width $2/\left\|w\right\|_2$.
Weak duality	$q(\alpha, \nu) \le f(x)$ for feasible x, $\alpha \ge 0$; so $d^\star \le p^\star$.
Slater	Convex problem with a strictly feasible point $\Rightarrow p^\star = d^\star$, dual attained. Affine constraints need only feasibility.
KKT	Stationarity, primal and dual feasibility, $\alpha_i^\star c_i(x^\star) = 0$. Necessary under strong duality; sufficient for convex problems.
SVM dual	$\max \mathbf{1}^\top \alpha - \frac{1}{2}\alpha^\top Q \alpha$, $Q_{ij} = y_i y_j K_{ij}$, $y^\top \alpha = 0$, $0 \le \alpha \le C$.
Recovery	$w = \sum_i \alpha_i y_i x_i$; $b = y_s - \sum_j \alpha_j y_j K_{js}$ for a free SV s.
Mercer	K valid $\iff$ every Gram matrix is SPSD; then $Q \succeq 0$.
SMO step	$\alpha_2 \leftarrow \mathrm{clip}_{[L,H]}\big(\alpha_2 + y_2(E_1 - E_2)/\eta\big)$, $\alpha_1 \leftarrow \alpha_1 + y_1 y_2(\alpha_2^{\mathrm{old}} - \alpha_2^{\mathrm{new}})$.
Kernel prediction	$f(x) = \sum_{i \in \mathrm{SV}} \alpha_i y_i K(x_i, x) + b$, $\hat{y} = \operatorname{sign} f(x)$; $w = \sum_i \alpha_i y_i \phi(x_i)$ is never formed.
One-vs-all	M SVMs (class k vs. rest); $\hat{y} = \arg\max_k f_k(x)$.
One-vs-one	$M(M-1)/2$ SVMs (class k vs. class l); $\hat{y}$ = class with most votes, ties to the smaller index.

4.13.1 Review questions

1. Write the lines $x_1 - x_2 = -2$ and $x_1 - x_2 = 4$ in the form $\langle w, x\rangle + b = \pm 1$ and compute their distance two ways.
2. Repeat Section 4.6 for the constraint $x - 3 \ge 0$. Find $q(\alpha)$, $\alpha^\star$, and verify all four KKT conditions.
3. In problem (B), the constraint is active but $\alpha^\star = 0$. Construct a one-dimensional problem in which the constraint is inactive, and explain why complementary slackness then forces $\alpha^\star = 0$.
4. Derive the soft-margin dual (4.16) again, but now explain why μ_i disappears and where the bound $\alpha_i \le C$ comes from.
5. Show that $K(x, z) = \langle x, z\rangle^2 + \cos\big(\angle(x, z)\big)$ is a valid kernel on nonzero vectors, using the closure rules and Proposition 4.10.
6. Redo Case 1 of the SMO example starting from the different feasible pair $\alpha_1^{\mathrm{old}} = 0.3$, $\alpha_2^{\mathrm{old}} = 0.8$ (still on the line $\alpha_1 - \alpha_2 = -0.5$). Compute E_1, E_2 and show that the update again gives $\alpha_2^{\mathrm{new}} = 0.7$. Why does the result not depend on the starting point on the line?
7. In the `quadprog` dual for D1, change C to 0.3. Which multipliers hit the upper bound, and what does the KKT table say about those points?
8. Explain why the primal QP needs no kernel but cannot use the Gaussian kernel, while the dual can.
9. In Example 3 of Section 4.11, classify $\tilde{x} = (0.5, -0.5)$. Compute the four kernel values and show that $f(\tilde{x}) \approx -0.416$.

10. Using (4.23), show that $\|w^\star\|^2 = \alpha^{\star\top} Q \alpha^\star$, so the margin $2/\|w^\star\|$ of a kernel SVM can be computed without ϕ.

11. In the three-class example, explain why the $\arg\min$ rule would label $\tilde{x}_1 = (1, 1.8)$ as class 2, and why that is wrong.

12. Find a point where one-vs-all predicts class 3 but one-vs-one predicts class 1, using the decision regions of the three-class example.

13. With $M = 10$ classes and 1000 points per class, how many binary problems and how many training points per problem does each strategy need?